

Pelagic: A Real-Time Mathematical Rendering System for Procedural Jellyfish, Stateful Appendages, Underwater Optics, and Interactive Liquid Glass

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Abstract

Real-time organism scenes become difficult to maintain when anatomy, motion, environmental effects, camera control and interface optics evolve as unrelated demonstrations. Pelagic is a browser-based rendering study that connects these systems while preserving an art-directed jellyfish identity. Its implementation combines a procedural mantle and folded oral membranes, pulse-coupled locomotion, persistent constrained appendages, bounded world-space currents, projected-size population detail, and an input-driven six-degree-of-freedom camera journey. A shared optical compositor samples one live ocean image for refractive bubbles, localized thermal shimmer and an interactive liquid clock. The clock combines authored implicit-surface choreography with a low-resolution persistent spring/advection field. This paper describes the actual Three.js 0.175.0 implementation, distinguishes code-equivalent equations from abstractions and artistic heuristics, and links the methods to reproducible source locations. The accompanying artifact includes browser motion evidence, deterministic tests, historical rejected approaches and fresh frame-interval measurements of the approved runtime. The work is a systems and visual-engineering study, not a fluid-structure interaction model, calibrated biological simulation or claim of algorithmic priority. Image-space optics cannot recover unseen radiance; liquid topology is not mass-conserving; performance and compatibility evidence remain specific to the tested hardware and backend.

1. Introduction

Pelagic presents a quiet underwater scene rather than a conventional portfolio of interface panels. Jellyfish move through open water, an input-driven camera visits several compositions, a brief bubble passage bends the live image, and a deep basalt sanctuary supplies a different scale of environmental detail. After inactivity, the same moving ocean becomes the source image for a liquid clock. These changes create a useful engineering problem: the visual identity must survive changes in scale, orientation, detail level and optical treatment.

A compelling still is insufficient. A rim can look continuous at rest yet break under contraction. A membrane may resemble tissue from the front but reveal a flat strip underneath. A distant animal can have a smooth silhouette while its appendages rotate rigidly with its body. A transparent clock can look luminous without bending anything beneath it. Pelagic’s development therefore relied on

controlled motion checks as well as attractive views. Figure 1 introduces the runtime; the motion supplement separates the behaviors that a still cannot prove.

The publication has a narrower claim than the ambition of the artwork. It documents one implemented system and its tradeoffs. It does not compare itself numerically with unrelated graphics demos, report user-study outcomes, or infer physical fidelity from resemblance. The authoritative runtime is commit `bce3571b0300ecfe5dc6e5dd45f28a9cda57006e`. Publication commits add explanation, measurement and packaging without changing that artwork. An unapproved later background experiment is discussed only as supplemental negative evidence.



Figure 1. Approved-runtime overview: luminous procedural tissue within the shared ocean. Actual browser capture; camera state and renderer metadata accompany the figure.

2. Contributions

The contribution is the documented integration and its inspectable artifact, not a new physical law. Five aspects organize the paper:

1. A shared anatomical surface contract for the bell, rolled margin and appendage roots, paired with folded membranes and stateful transport.
2. A pulse state reused for deformation and propulsion, observed by a bounded current/wake system rather than replaced by an independent decorative halo.

3. Population detail that follows projected importance while retaining the approved tissue implementation and persistent appendage state.
4. One live-ocean optical source shared by bubbles, thermal shimmer and idle glass, with explicit approximation boundaries and resource ownership.
5. A reproducible account of source-derived mathematics, motion checks, performance, rejected designs and remaining limitations.

Each is an engineering result. The equations are deliberately modest: familiar building blocks become useful when their coordinate spaces, timing and ownership are made consistent. Table 1 identifies those responsibilities.

3. Related work and scope of influence

The biological literature distinguishes mechanisms that a stylized animation can easily collapse into a single sinusoid. Costello and colleagues review jellyfish swimming across morphological and hydrodynamic regimes (Costello et al. 2021). Gemmell and colleagues describe passive energy recapture during refill (Gemmell et al. 2013). These sources motivate a distinction between contraction, recovery and coast. Pelagic does not reproduce their experimental apparatus, pressure fields, efficiency measurements or species-specific kinematics. Its secondary thrust is an authored temporal signal, not an estimate of recovered energy.

Position-based dynamics provides relevant context for correcting geometric constraints after a prediction step (Müller et al. 2007). Pelagic’s appendages use previous positions, damping and repeated length projection. They do not implement the complete general method, derive compliance from tissue measurements, or solve a coupled elastic volume. A small oral-spine separation operation is a visual contact heuristic, not a comprehensive self-collision system.

Blinn’s implicit surfaces establish the usefulness of summing smooth density functions to form connected shapes (Blinn 1982). Pelagic applies additive implicit contributions to a two-dimensional clock field. No three-dimensional isosurface is extracted, and the apparent liquid volume is not conserved. Image-space refraction work, including Wyman’s two-interface approximation (Wyman 2005), supplies a useful conceptual boundary: plausible optical images can be produced without complete scene transport, but missing image information remains missing. Pelagic solves analytic ellipsoid interfaces and samples its current ocean image; it is not a reproduction of Wyman’s entire pipeline.

The software foundation includes an adapted MIT-licensed Aurelia scene (Niehus, n.d.) and the pinned Three.js renderer (Three.js contributors 2025). FluidGlass was studied as an interaction reference (chiuhans111, n.d.), not copied as source, shader, layout or media. These implementation and design references are not scientific validation. Third-party provenance and retained notices accompany the artifact.

4. System overview

React owns readiness, idle intent and sparse semantic controls. High-frequency work belongs to imperative scene objects and persistent buffers. The scene uses `WebGPURenderer` from `three/webgpu`, with an explicitly verified WebGL2 backend for this publication. A package dependency on React

Three Fiber does not make this an R3F renderer: no Fiber scene ownership is used on the active path.

The frame reads input and updates bounded journey progress. The school director advances animal poses; the environment advances its persistent populations; the detail adapter evaluates screen importance. Connected-ocean state observes the animals and updates particulate. Tissue geometry and presentation then update, followed by the existing scene render. When optical effects are active, that render targets a clean color image and a shared output pass composes the effects. Idle adds small field-simulation passes, not another ocean.

Table 1. Active responsibilities at the publication runtime.

Responsibility	Owner	Persistent state
Browser/scene lifecycle	<code>HeroScene</code> , <code>useIdleScreen</code>	renderer, scene, idle intent, readiness
Main route poses	<code>JellySchoolDirector</code>	position, velocity, heading, pulse
Tissue	<code>LivingAppendages</code> , <code>PopulationAnimal</code>	chains, geometry, material uniforms
Detail assignment	<code>PopulationDetail</code>	detail state, pooled light/fleck ownership
Connected water	<code>CurrentField</code> , <code>ConnectedOcean</code> , <code>OceanSnow</code>	event pools and particle positions
Camera	<code>JourneyController</code> , <code>TrackPlayer</code> , <code>ViewController</code>	scalar progress, baked poses, Explore state
Optical output	<code>LiveOceanLens</code>	clean scene target and output material
Sanctuary	<code>Sanctuary</code> , <code>VentDynamics</code>	geology resources, plume, pooled illumination
Idle	<code>OceanIdleGlass</code> , <code>IdleDisplacement</code>	two masks, two field targets, bounded choreography

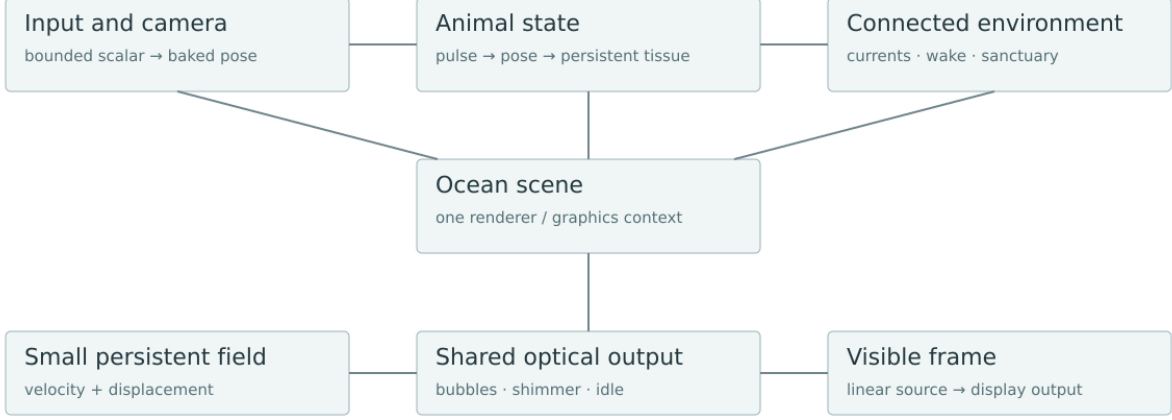
Figure 2 summarizes the ownership boundaries and the convergence on one ocean image.

5. Notation and spaces

World coordinates are right-handed with Y up. The bell-first animal axis is local +Y, with X/Z spanning the rim. Appendage positions are animal-local, but their previous state is transported to preserve world-relative lag. Camera view direction is $-Z$. Output UV is top-origin; the projection code explicitly inverts clip Y. Canvas clock-mask sampling performs its own Y inversion.

Time is in seconds, except normalized pulse phase q and journey progress s . World lengths are artistic units, not meters. h denotes a fixed simulation step. The mantle uses latitude parameter τ and azimuth θ ; u is a transverse membrane coordinate. $S(x)$ denotes cubic smoothstep after clamping x to $[0, 1]$. The complete symbol table is in `notation.md`.

One world, separate responsibilities



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Figure 2. Ownership and frame composition. Auxiliary field/output draws are not additional ocean scene renders.

An equation labeled **CODE-EQUIVALENT** matches the stated source subexpression, including its constants. It need not describe every surrounding branch. A **CONTINUOUS ABSTRACTION** summarizes an algorithm without claiming bitwise equivalence. An **ART-DIRECTION HEURISTIC** expresses a designed relationship, not a physical inference. These labels describe different questions from correctness: an exact transcription can still be physically approximate. `source-map.md` records immutable paths and line ranges for every equation.

6. Procedural jellyfish geometry

6.1 Mantle and root continuity

The bell and rim share `mantlePoint`. Let R, H be pulse-adjusted radius and height, p the bell contraction, L the lobe count, y_r the rim offset and m_r the margin-roll signal. Let c_x, c_y be the two-component local surface current. With $m = \tau^5$ and $f = \cos(L\theta)$, equation (1) is **CODE-EQUIVALENT**:

$$\begin{aligned}\rho &= \sin(\pi\tau/2)R(1 - 0.07pm)(1 + 0.012fm), \\ x &= \cos\theta\rho + 0.08c_x\tau^2, \\ y &= \cos(\pi\tau/2)H + y_r\tau^2 + m(0.014f + 0.04m_r), \\ z &= \sin\theta\rho + 0.08c_y\tau^2.\end{aligned}$$

$\tau = 0$ is the apex; $\tau = 1$ is the equatorial region; the surface extends to a softly rolled margin near 1.12. Rim tentacles attach at 1.1 on this same function. This avoids tuning disconnected rim geometry against the bell. Periodicity in θ is maintained while the bell contracts. Seam normals and fractional-power domains require separate numerical care: a valid surface does not prevent a

shader NaN from appearing as a black seam.

6.2 Folded oral membranes

An oral arm is not merely a wide line. Its simulated spine carries a transverse folded section with narrowed insertion and rounded tip. For normalized length $t \in [0, 1]$, transverse $u \in [-1, 1]$, arm index a , and animation phase ϕ , equation (2) is **CODE-EQUIVALENT**:

$$\begin{aligned} w &= (0.055 + 0.32 \sin(\pi t)^{0.65}) \sqrt{\max(0, 1 - t^8)}, \\ b &= uw[1 + 0.20|u| \sin(30t - a)], \\ f &= [\sin(2\pi u + 7t + a) - \sin(7t + a)] 0.30w \\ &\quad + 0.32w|u|^2 \sin(30t + 2u - a) + 0.04w|u| \sin(22t - 0.8a - 0.42\phi). \end{aligned}$$

The section is reconstructed in a frame along the spine. The subtraction in the first fold term keeps the central spine on its simulated centerline. Free-edge terms add folding without moving the whole arm as an independent object. The procedure remains a surface approximation: intersections can occur because folded sheets have no volumetric collision solver.

Figure 3 plots the mantle profile and transverse membrane sections directly from these functions.

Continuous mantle and folded membrane



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Figure 3. Mantle coordinates and folded section, plotted from the approved functions rather than traced from a screenshot.

7. Bell kinematics

The normalized swim phase wraps continuously. Contraction occupies the first 0.20 of the cycle, re-fill continues to 0.68, and the remaining interval is a coast. Equation (3), **CODE-EQUIVALENT**, defines contraction magnitude:

$$p(q) = \begin{cases} S(q/0.20), & q < 0.20, \\ 1 - [S((q - 0.20)/0.48)]^{1.42}, & 0.20 \leq q < 0.68, \\ 0, & q \geq 0.68. \end{cases}$$

Define a window $W(q; a, b) = \sin^2[\pi(q-a)/(b-a)]$ strictly inside (a, b) and zero elsewhere. Equation (4), **CODE-EQUIVALENT**, gives primary thrust $T_1 = W(q; 0.025, 0.215)$, secondary thrust $T_2 = W(q; 0.57, 0.82)$, and margin roll $m_r = 0.78T_1 - 0.52W(q; 0.22, 0.66)$.

The same state reaches geometry and locomotion. It produces a temporal relationship, not a claim of measured pressure. Activation adds an integrated response through the existing tissue state rather than creating a replacement animal. Figure 4 plots the exact signals; S1 supplies the corresponding motion; S1 shows several cycles without converting timing into a still-image claim.

8. Locomotion and schooling

Each main-route animal has a desired location derived from an authored path plus slow current-like variation. Its position is integrated from velocity; it is not simply placed on that desired point each frame. The direction blends toward the route with a contraction-dependent gain. In equation (5), **CODE-EQUIVALENT** for the velocity core, $\hat{d}$ is the updated swimming direction, $e = X_{desired} - X$, $k = \min(0.42, 0.075 + 0.045\|e\|)$, $a = (0.72 + 0.28 \text{ scale}) \text{ motionScale}$, and c is the two-dimensional current:

$$\begin{aligned} V^* &= V + h[(2.35T_1 + 0.58T_2)a\hat{d} + ke + 0.095 \text{ motionScale}(c_x, 0, c_y)], \\ V^+ &= \text{cap}_{v_{max}} \{V^* e^{-h(0.24 + 0.22 \text{ refill} - 0.08 \text{ coast})}\}, \\ X^+ &= X + hV^+. \end{aligned}$$

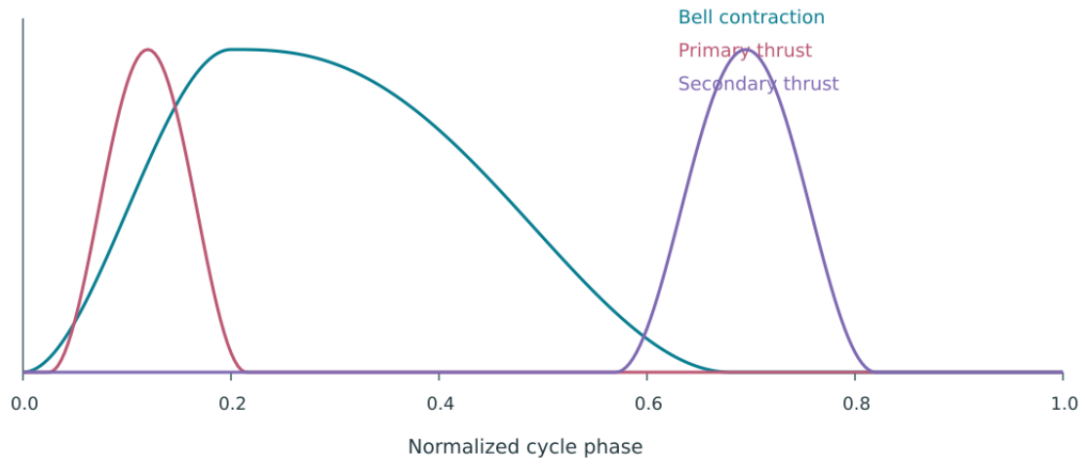
The cap is $v_{max} = 0.92 + 0.42 \text{ drift}$. A separate soft composition correction engages when route error exceeds 4.2 world units; excluding it would overstate the purity of propulsion-driven travel. Orientation follows velocity with a pulse-dependent quaternion blend. No calibrated body drag or added-mass model is implied by the exponential factor.

For pairs of visible main-route animals, equation (6), **CODE-EQUIVALENT**, adds opposite desired-position corrections when $d < d_{safe}$:

$$C_{ij} = 0.44(1 - d/d_{safe}) \frac{X_{desired,i} - X_{desired,j}}{d}, \quad d_{safe} = 1.05 + 0.55(\text{scale}_i + \text{scale}_j).$$

Distance is clamped below at 0.001. This steers desired routes apart; it does not resolve collisions between all membranes or represent a general schooling model. The distant route source has its own established motion history; the population adapter shares rendering quality, not a newly unified route solver.

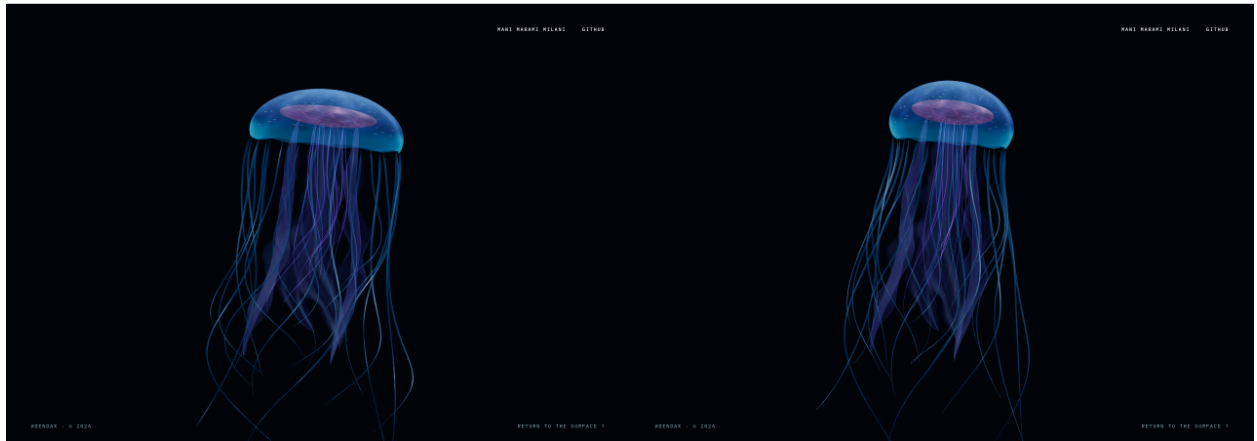
Unequal contraction, refill and coast



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$q=0.059 \cdot t=9.4 \text{ s}$

$q=0.209 \cdot t=10.1 \text{ s}$



$q=0.509 \cdot t=11.5 \text{ s}$

$q=0.809 \cdot t=12.9 \text{ s}$

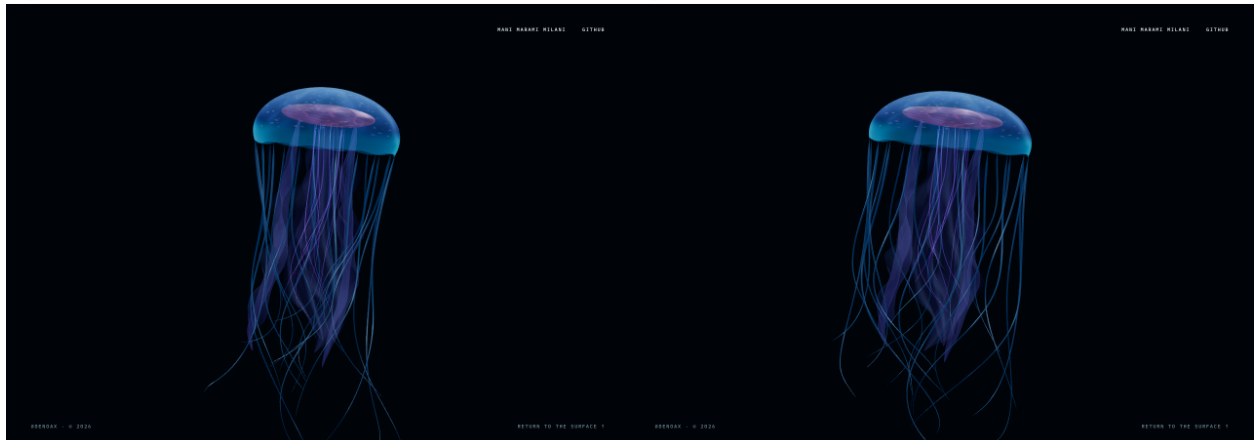


Figure 4. Exact source-derived pulse curves and four actual browser poses with recorded phase q and simulation time. Fixed specimen view and distance; contraction, refill and coast are unequal. S1 supplies continuous motion.

Algorithm 1 — animal update (structural pseudocode).

```
evaluate authored desired route and bounded drift
accumulate pairwise desired-route separation
sample contraction/refill/coast state
turn swimming direction toward desired route
integrate pulse thrust, weak tether and current into velocity
apply drag and speed cap; integrate position
apply exceptional soft composition guard if far outside route
orient bell-first axis toward velocity; preserve physical scale
publish pose and pulse for tissue and environmental observers
```

9. Stateful appendages

The appendage history is meaningful only if it survives body movement. Before local simulation, current and previous points are rotated by the change from the previous body frame and translated by the same inverse body displacement. Applying a different transport to previous state would inject spurious velocity. Large discontinuous body displacements are guarded instead of replayed as a violent chain impulse.

Equation (7) is **CODE-EQUIVALENT** for prediction. With σ the bounded frame scale and $d_0 = 0.955$ for arms or 0.968 for tentacles,

$$\tilde{x}_i = x_i + d_0^\sigma(x_i - x_i^-) + g_i + c_i + e_i + r_i.$$

The named increments are not a single physically measured acceleration: g_i is downward bias proportional to σ^2 ; c_i is current displacement proportional to longitudinal $t_i^2\sigma^2$; e_i is bounded traveling flow; r_i is a localized pointer repulsion. The approved path uses $h = 1/60$, giving $\sigma = 1$ during fixed simulation steps. Replacing these mixed increments by an unexplained h^2F would misstate the implementation.

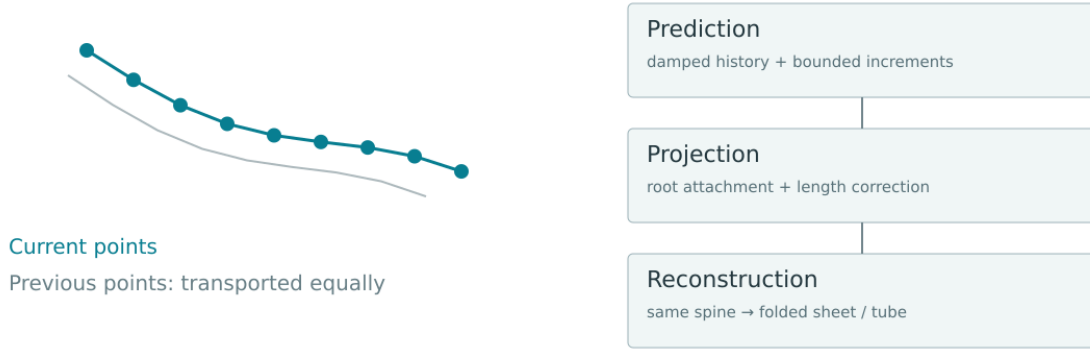
Roots are reattached during projection. For segment vector $\delta = x_i - x_{i-1}$, $\ell = \max(10^{-4}, \|\delta\|)$ and rest length r , equation (8), **CODE-EQUIVALENT**, is

$$\epsilon = (\ell - r)/\ell, \quad x_{i-1} \leftarrow x_{i-1} + 0.46\epsilon\delta, \quad x_i \leftarrow x_i - 0.54\epsilon\delta.$$

The first free point instead receives the full correction because its parent is fixed. Arms use five iterations, tentacles four. Three oral-spine separation passes inspect neighboring longitudinal samples and apply the same bounded correction to both Verlet histories. This reduces some oblique crossings without adding artificial propulsion; it is not sheet-sheet collision.

Visible tubes use frames transported along the chains. Folded arm surfaces use the simulated centerline, not independent animation transforms. Normal refresh and deformation scheduling depend on significance; detail transitions resample the same state. S2 shows turning and lag. Figure 5 distinguishes state from reconstructed surface so a smooth mesh is not mistaken for a denser solver.

History belongs to the appendage, not the frame



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Figure 5. Persistent chain history, length projection and reconstructed membrane/tube surfaces.

Algorithm 2 — appendage step.

```

transport both position histories into the current animal frame
accumulate bounded time; take fixed tissue steps
for each active chain:
    attach root to approved anatomy
    predict free points from damped history and authored increments
    repeat length projection; reattach root
separate nearby oral spines, correcting both histories equally
resample existing chains at the visible detail level
reconstruct folded membranes and tapered tubes; refresh needed normals

```

10. Tissue optics and bioluminescence

The bell uses `MeshPhysicalNodeMaterial`, but its translucent identity is an explicit approximation. The material sets transmission to zero: it does not refract the ocean framebuffer. Anatomical thickness, facing and absorption control alpha, pigmentation and roughness. This distinction matters because a living-looking bell is not evidence of physical subsurface transport.

For clamped UV latitude v , let K be the canal mask and $f = \max(0.24, |n_v \cdot v_{eye}|)$. Equation (9), **CODE-EQUIVALENT** for the bell path, is

$$\begin{aligned}
 d_t &= 0.30(1 - v)^{1.4} + 0.04 + 0.20K \max(0, \sin \pi v), \\
 A &= 1 - e^{-1.7d_t/f}, \\
 \alpha &= \text{clamp}_{[0.025, 0.62]}[(0.40A + 0.08) \text{opacity}/0.58].
 \end{aligned}$$

The membrane branch uses a different thickness/edge profile. UV clamps protect fractional powers

against tiny interpolation overshoots. This is an important example of a defect that could resemble geometry segmentation while originating in material evaluation.

A seeded 512×256 data texture stores soft mottling, canal structure and sparse luminous cells in separate channels. It includes 112 seeded cell marks and matching seam texels, with mipmapping for distance. Emission has a hierarchy: soft tissue, brighter rim, pink/violet internal accents and restrained speckles. A broad analytic environment lobe provides a moving highlight as normals turn. Bloom is not used to construct the animal; the inherited bloom chain is off.

The colors are art direction, not species spectroscopy. Double-sided alpha surfaces can still sort imperfectly when they overlap. Tissue thickness here is a shading profile, not the distance between a watertight inner and outer shell. These limits remain visible in close inspection and are not hidden by calling the material physically accurate.

11. Connected current and wake

The current system is inexpensive enough to sample for multiple CPU particle systems. Let $t' = 0.075t$. Equation (10), **CODE-EQUIVALENT**, defines ambient flow:

$$\begin{aligned} U_x &= 0.055 + 0.085 \sin(0.21y + t') + 0.045 \cos(0.17z - t'), \\ U_y &= 0.025 + 0.055 \sin(0.19z + 0.7t') + 0.035 \cos(0.18x + t'), \\ U_z &= 0.07 \sin(0.20x - 0.8t') + 0.04 \cos(0.23y + t'). \end{aligned}$$

Each component is independent of its own coordinate, so this ambient expression has zero divergence analytically. The complete field does not inherit that guarantee: localized envelopes and the final velocity cap alter it.

For wake age a , lifetime L , source axis $\hat{b}$, offset $r = X - X_e$, initial radius R_0 and strength k , set $R = R_0(1 + 0.08a)$ and $d^2 = \|r\|^2/R^2$. Equation (11), **CODE-EQUIVALENT**, applies inside $d^2 < 4$:

$$E = S(1 - d^2/4)^2 S(a/0.3) S((L - a)/3) k, \quad \Delta U = 0.31 E (\hat{b} \times r) / R - 0.23 E \hat{b}.$$

There are 32 wake slots and eight activation slots. Wake emission observes an actual rising pulse threshold and bounded animal speed; it does not attach a permanent cloud to the body. Events drift downstream, expand modestly and die. Repeated activation is rate-limited per source, and a single delayed neighbor response is non-recursive. The adapter observes approved animal state rather than retuning its physics.

Marine snow uses three world-space layers: 64 near, 320 mid and 400 distant particles on desktop, with distinct size, opacity, extent and near-distance envelopes. Particle motion samples the common field plus a small settling bias. Recycling occurs beyond a zero-opacity boundary and fades back in. A separate bounded fleck pool follows the same field. S3 shows the local activation response and decay; sparse highlights must remain subordinate to the animals.

12. Projected-importance population detail

Detail follows CSS-pixel bell diameter, not animal identity or drawing-buffer DPR. For radius R , positive view depth z , viewport height H , camera zoom Z and vertical field of view f , equation (12), **CODE-EQUIVALENT**, is

$$D = \min \left(8H, \frac{RHZ}{\max(R/4, z) \tan(f\pi/360)} \right), \quad d^+ = d + \text{sgn}(k - d) \min(|k - d|, \min(h, 0.05)/1.2).$$

Invalid inputs return zero coverage; the second expression advances continuous detail toward integer target tier k , with no hidden-tab debt. Table 2 records the hysteresis. Conservative bounds include appendages when the bell is clipped.

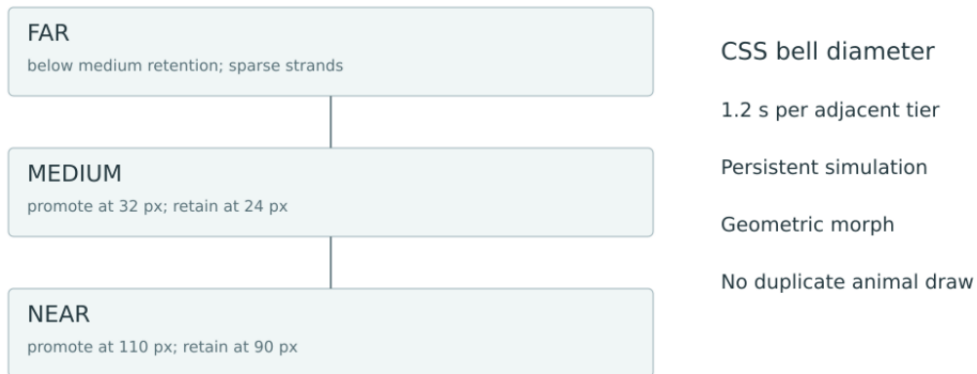
Table 2. Selected source constants, not measured biological parameters.

Quantity	Value	Meaning
Near promotion / retention	110 / 90 CSS px	avoids repeated threshold toggling
Medium promotion / retention	32 / 24 CSS px	same hysteresis principle
Interaction coverage	24 CSS px	plus visibility and presence >0.15
Adjacent detail travel	1.2 s	reversible geometric transition
Tissue / current / idle field step	1/60 s	distinct subsystem accumulators
Journey step	1/240 s	bounded input follower
Wake / activation lifetime	5.8 / 6.4 s	finite disturbance ownership
Optical heroes / ambient bubbles	5 / 384	not 389 full refraction evaluations
Idle field long dimension	256 samples	each dimension at least 64

Lower levels resample the same mantle and persistent oral spines. Resources and tube indices are prepared in advance. During a transition, neighboring surface samples morph geometrically; this is not two complete transparent animals alpha-crossfading. Selected tentacles taper in by weight, with dormant histories seeded from a live neighboring strand. Near uses the approved implementation. Offscreen surface reconstruction is skipped while state needed for return is preserved.

Figure 6 connects the threshold policy to recorded population assignments at three journey positions.

Importance with hysteresis and persistent state



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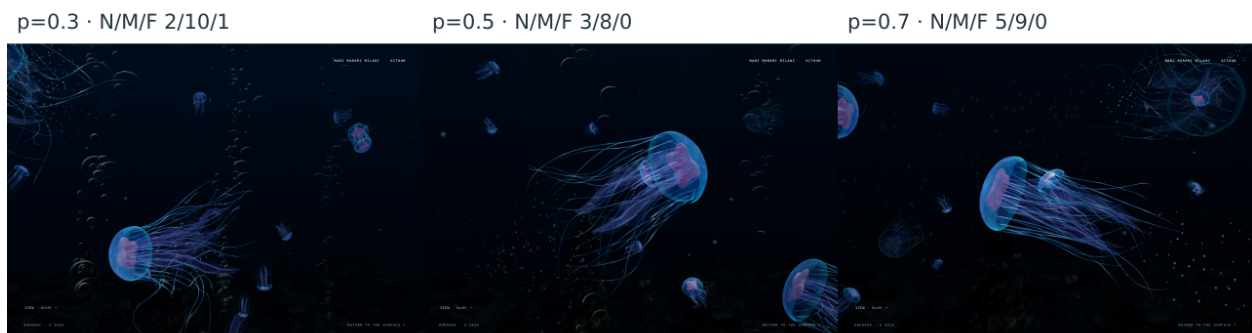


Figure 6. Projected-size detail thresholds and actual population snapshots under identical view-port/DPR settings. N/M/F are near/medium/far counts. Different journey positions are not a forced same-pose tier ablation. CSS thresholds do not change with DPR.

Algorithm 3 — importance detail.

```
read existing main and distant route poses
test conservative animal bounds against camera frustum
compute CSS bell diameter; choose hysteretic target tier
advance continuous detail without pause debt
reuse persistent spines; sample adjacent prepared surface levels
morph geometry and strand widths, not two full animal renderings
assign finite light/fleck resources by visible importance
retain normal activation eligibility and approved near presentation
```

13. Camera journey and exploration

The camera is input-driven. Wheel line/page units normalize to CSS-pixel intent; touch and keyboard inputs reach the same bounded follower. Opposite intent cancels queued travel without teleporting the camera. No production autoplay advances the journey when the visitor stops.

With progress error $e = s_{target} - s$, gain g , speed bound $v_m = 0.14$, acceleration bound $a_m = 1.4$ and $h = 1/240$, equation (13), **CODE-EQUIVALENT**, is

$$\begin{aligned}v_d &= \text{sgn}(e) \min[v_m \tanh(g|e|), \sqrt{2a_m|e|}], \\v^+ &= v + \text{clamp}(v_d - v, -a_m h, a_m h), \\s^+ &= s + h v^+.\end{aligned}$$

The target stays at most 0.075 ahead/behind progress and within $[0, 1]$. Balanced gain is 72; cinematic/responsive use 52/92. Tiny residual motion snaps to exact rest only within the allowed braking step. A legacy **ProgressSpring** class remains in source but does not define current production travel.

Each view bakes 1,201 poses. For a scalar coordinate over one authored segment, let p be the starting value, Δ its difference, and m, n the endpoint tangents multiplied by segment span. Equation (14), **CODE-EQUIVALENT**, is

$$P(t) = p + mt + (10\Delta - 6m - 4n)t^3 + (-15\Delta + 8m + 7n)t^4 + (6\Delta - 3m - 3n)t^5.$$

Interior tangents use a monotone harmonic construction; endpoint tangents are zero. Authoring angles unwrap before conversion to hemisphere-continuous quaternions. Runtime sampling linearly interpolates position and spherically interpolates quaternion. Thus the authoring polynomial is smooth, while the finite sampled playback is an approximation to it—not an analytic continuous quaternion spline.

Documentary, Drift, Intimate and Deep expose different authored tracks. Explore temporarily gives the visitor observer control without rebuilding the ocean. Idle holds the observer and journey while animals continue moving. S6 shows view changes and Explore; the paper does not reinterpret those controls as simulation steering or a new camera design.

14. Shared optical compositor

One ocean image supplies all active optical effects. `LiveOceanLens` owns a full drawing-buffer RGBA16F target, a depth attachment, and the output material. The scene is rendered into that target without the lens in its input. Thermal shimmer and bubble slots sample that clean source; idle may then replace local output with its own clean-source lookup. There is no recursive color feedback.

This ordering has a cost and a limitation. The output adds a screen-sized pass when optical content is active; idle also advances its small persistent field. However, overlapping optical effects do not solve nested ray transport. Each effect sees the same unmodified ocean source. The blend can preserve an attractive image without representing light that physically traversed every intervening surface.

The target uses renderer sample count and linear color. Tone/output state is restored around preparation and rendering, including failure paths. A tiny scratch target is used during idle preparation and then disposed. Ready state is published after a valid first optical draw, not merely after allocating a material. Figure 7 diagrams source versus output ownership.

Clean source is shared, never recursively overwritten



Field targets store deformation only. All refractive lookups read the clean ocean color.

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Figure 7. Shared clean-image compositor. The output is never sampled as its own input; idle field ping-pong stores state, not ocean color.

Algorithm 4 — optical frame.

```
advance bounded idle field, if needed
prepare active thermal/bubble domains
if no optical domain and no visible idle: render ocean directly
otherwise:
    resize clean target to actual drawing buffer
    render the real ocean once into clean target
    sample clean image for thermal and bounded bubble slots
    if idle visible: evaluate liquid field and clean-image displacement
    render output once; restore renderer state even after errors
```

15. Bubble optics and lifetime

For lens-local origin o , direction d and ellipsoid radii vector r , define $O = o/r$, $D = d/r$. Equation (15), **CODE-EQUIVALENT**, is

$$\begin{aligned} A &= D \cdot D, \quad B = O \cdot D, \quad C = O \cdot O - 1, \\ t_e &= \frac{-B - \sqrt{\max(B^2 - AC, 10^{-6})}}{A}, \\ n &= \text{normalize}[(o + t_e d)/r^2]. \end{aligned}$$

The shader transforms the normal by the appropriate normal matrix and refracts in metric view space. A second analytic intersection finds the exit; a second refraction computes the outgoing direction. This is different from deforming only a visible lens mesh while leaving the source image unchanged.

Equation (16), **CODE-EQUIVALENT** for the projection core, uses exit point E_v , outgoing view ray D_o , scene view-depth z_s , and camera projection P :

$$\begin{aligned} z_i &= \max(z_s, E_{v,z} - 6), \\ \ell &= \max\left(0, \frac{z_i - E_{v,z}}{\min(D_{o,z}, -0.05)}\right), \\ q &= P(E_v + \ell D_o, 1), \\ uv_r &= (q_x/(2q_w) + 1/2, 1/2 - q_y/(2q_w)). \end{aligned}$$

The actual code guards the homogeneous denominator, grazing rays, source borders and depth tests. Bubble lookup displacement is capped to $\sqrt{2} 0.008$ in UV length before further masks. The virtual image plane six units behind the exit is explicitly approximate, especially for transparent animals that do not write depth. A depth texture is not translucent depth, and its attachment is not proof that the selected backend resolves it correctly. The pinned WebGL multisample limitation is recorded in the supplement.

Five refractive heroes accompany 384 cheap ambient films. Seeded lifetimes, stream packets, aspect-changing shape and modest drift make a bounded passage, not a permanent screen of lenses. Only heroes perform the analytic optical lookup. Authored index ratios are softened rather than using a

lateral water/air ratio that would require unavailable reflected scene rays at grazing angles. Figure 8 and S5 expose the image bending and its limitations in motion.

16. Abyssal sanctuary and participating-media approximation

The deep scene uses a dark basalt basin, connected shelves, a recessed channel, one principal sulfide structure, secondary spires and sparse vent-associated life. These are procedural and instanced artistic forms, not a surveyed site. Processed Rock 07 scan luminance ([Poly Haven, n.d.](#)) contributes mineral variation. The active geology material is `MeshBasicNodeMaterial` with an authored surface-gradient lighting response; calling it a calibrated PBR BRDF would be incorrect. Scanned detail and physically motivated placement do not change that.

The plume maintains fixed-capacity position, velocity, age, lifetime and size arrays. It is pre-established at construction rather than emitted as a first-visit burst. Nearby mineral packets share spatially related eddies. In equation (17), **CODE-EQUIVALENT** for selected updates, sampled current U is filtered into U_f , then a target plume velocity V_d is followed:

$$\begin{aligned} U_f^+ &= U_f + [\text{clamp}(U, -0.65, 0.65) - U_f](1 - e^{-h/0.85}), \\ V^+ &= V + (V_d - V)(1 - e^{-kh}), \quad X^+ = X + hV^+. \end{aligned}$$

$k = 2.1$ for smoke and 1.2 otherwise. Smoke’s vertical target includes $1.6e^{-0.18a} + 0.045$ plus filtered vertical current. Lateral entrainment grows with height. These are **ART-DIRECTION HEURISTICS** when interpreted as plume physics: no temperature, pressure or buoyancy conservation is solved.

Localized thermal domains reuse the optical output. Their world-space pattern and bounded projection produce shimmer without a second scene render. A single pooled animal-derived lighting contribution changes owner through zero rather than sliding an unrelated bright point across the floor. Channel-dependent distance attenuation blends toward directional water radiance. This is not volumetric multiple scattering or measured ocean image formation.

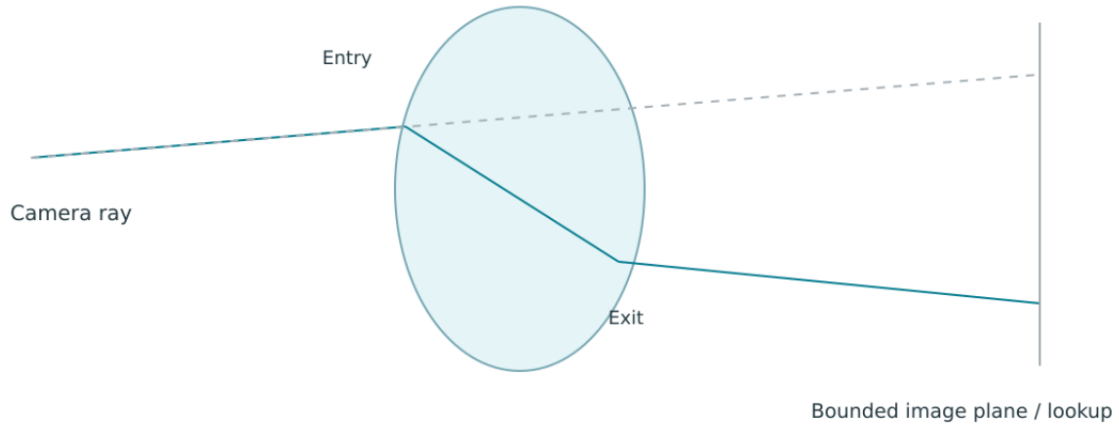
Figure 9 shows the resulting sanctuary within the approved ocean rather than an isolated geology renderer.

17. Idle state and implicit liquid topology

Idle is a mode of the shared compositor, not another WebGL canvas. The semantic `IdleScreen` contains no renderer. A 30-second inactivity policy requests entry; readiness and reduced-motion conditions govern whether it appears. Pointer motion deforms the active film without dismissing it. Click/tap or the defined keyboard actions initiate dismissal and consume the event before it becomes an animal click. The ocean continues beneath a held camera pose.

The clock uses two independent canvas textures. Clone-sharing a texture source would not supply independent old/new masks in this Three.js version. Digits occupy fixed cells while retaining the font’s proportional shapes. Landscape uses HH:MM; portrait stacks hours and minutes. Eight contact sites are found from the project’s own low-resolution clock mask, not from ocean readback.

Analytic interfaces, approximate scene information



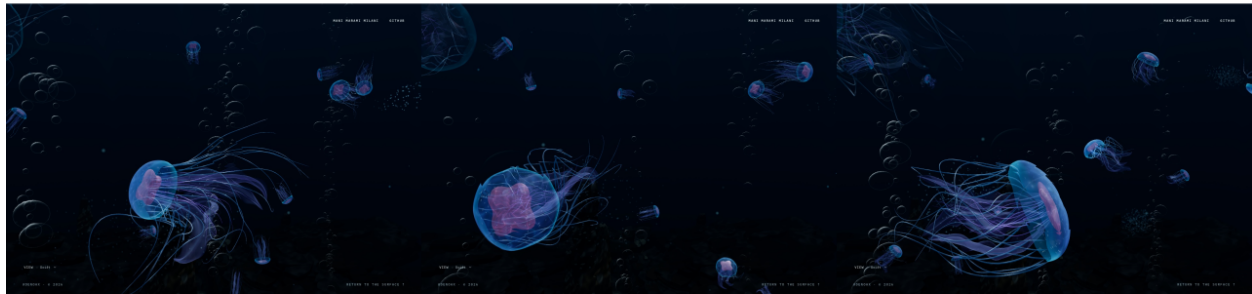
Schematic only: normals and directions are transformed in metric view space.

PELAGIC · source-derived diagram · approved M7 bce3571

S5 · source frame 110

S5 · source frame 230

S5 · source frame 360



Frozen live ocean: optics ON

Same frame state: optics OFF

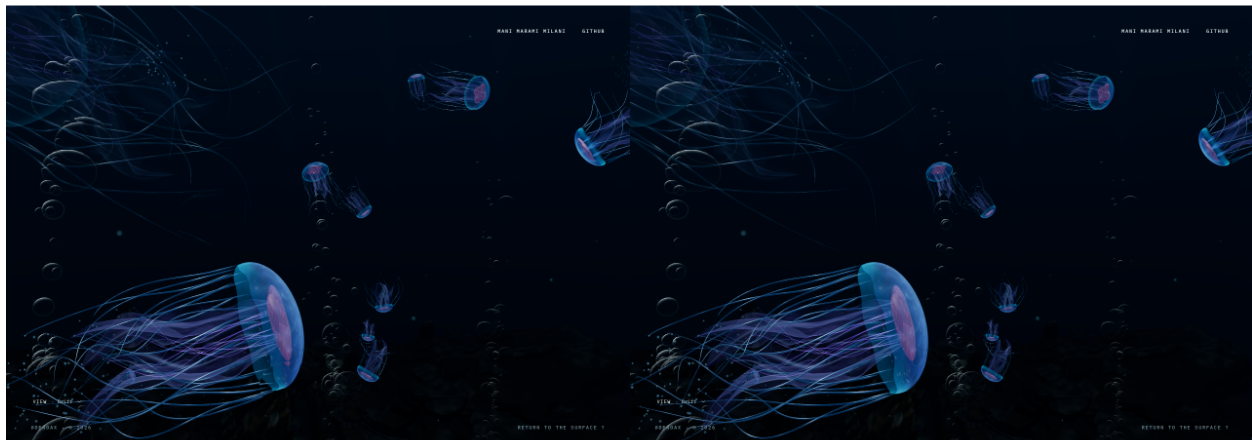


Figure 8. Analytic-interface schematic, three actual S5 browser frames and a frozen optics-on/off diagnostic. The virtual image plane is approximate; disabling hero refraction through the existing review API retains ambient bubble films. These are live-color distortions, not scene reconstruction.

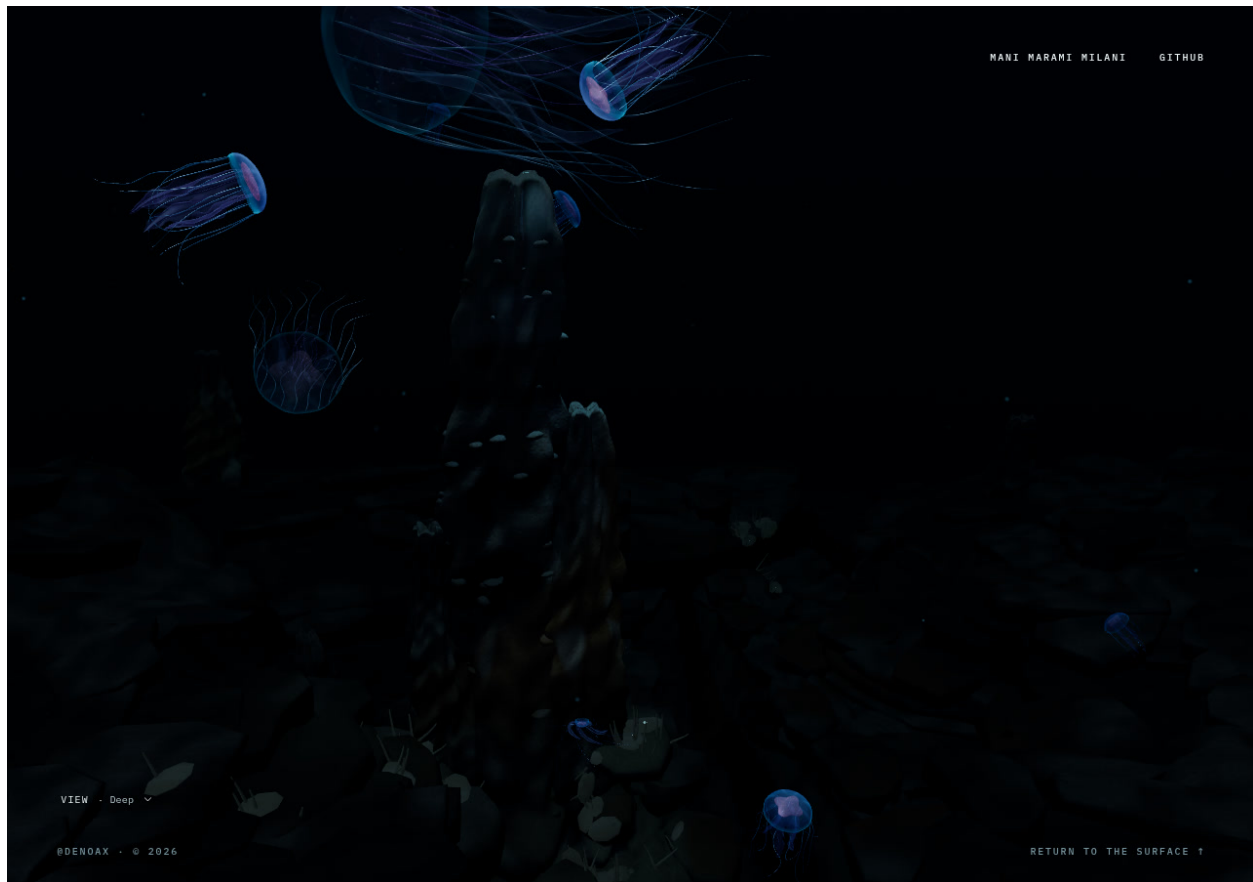


Figure 9. Approved sanctuary at the publication runtime: procedural geology, localized plume and restrained life.

Equation (18) is a **CONTINUOUS ABSTRACTION** of the scalar topology:

$$F(u) = a S_{0.12,0.95} \left(G(u - \Delta) + \sum_j D_j(u - \Delta) + \sum_j N_j(u - \Delta) + \sum_j R_j(u - \Delta) \right).$$

a is entry/exit amount; G is the mapped, locally grown/eroded glyph; D_j, N_j, R_j are droplet, neck and reservoir/bulge contributions. The actual function includes conditionals for changed cells, colon growth, front arrival, old/new stroke mapping and exit. It is not accurately represented by simply crossfading two text images. Gaussian-like additive masses create connected boundaries, but their sum is not a material-volume conservation law.

The field is evaluated at the sample and four offsets. In equation (19), **CODE-EQUIVALENT**, $e = (1.7/W, 1.7/H)$ and

$$\begin{aligned} g &= (F(u + (e_x, 0)) - F(u - (e_x, 0)), F(u + (0, e_y)) - F(u - (0, e_y))), \\ n &= \text{normalize}(-16g_x, -16g_y, 1), \\ \delta u &= 0.035 n_{xy}(H/W, 1) F(u) S(a) S_{0,0.06}(\text{border}). \end{aligned}$$

The difference is intentionally not divided by $2e$. Its scale and the factor 16 form an artistic normal construction. The resulting lookup samples the current clean ocean color at clamped $u + \delta u$. Small normal-driven light and luminance adaptation support the material; they do not substitute for live image displacement. S8–S12 show formation, contact, interaction, minute change and dismissal at normal presentation size.

Figure 10 samples contact and formation from S9; its complete motion remains the stronger temporal evidence.

18. Persistent deformation and event choreography

The paired field stores velocity in RG and displacement in BA, signed around zero in RGBA16F. Its long dimension is 256, with a minimum of 64 samples on either axis. It is small state, not a downsampled fake ocean. Current ocean color stays in the separate full-resolution optical target.

For screen coordinate u , first read previous velocity V_0 , backtrace $p = u - 0.42hV_0$, and sample state (V, D) plus its four-neighbor average $(\bar{V}, \bar{D})$. Equation (20), **CODE-EQUIVALENT**, gives the core step:

$$\begin{aligned} V' &= \text{clamp}_{[-0.58, 0.58]} \{ [0.88V + 0.12\bar{V} + h(36(\bar{D} - D) - 5.5D)] e^{-2.8h} + 3.2h V_p B I \}, \\ D' &= \text{clamp}_{[-0.17, 0.17]} (D + hV'). \end{aligned}$$

V_p is bounded pointer velocity, B a Gaussian segment brush, I fresh-input presence. A queued bounded event impulse is added to V' after its clamp and before displacement integration; an edge envelope multiplies the final state. That order is material: the first clamp alone is not an absolute bound after the event term. The method is a spring/advection construction, not incompressible Navier–Stokes, and no pressure projection is performed.



Figure 10. Normal-size clock contact and merge sequence. The supplement records actual timing and warns against interpreting the authored neck as conserved fluid.

Entry contact follows the facing mask surface, not an interior stroke anchor. The existing M7.2 choreography gives the approaching surfaces a visible gap, creates a narrow connection, then transfers and settles. Let $b = S(\text{age}/0.65)$ and $T = S((t - 0.29)/0.58)$ after the per-site contact time. Equation (21), **CODE-EQUIVALENT** for entry radius, is

$$r = (0.034 + 0.003(i \bmod 3))\sqrt{(1 - T)b}.$$

The source's scalar `mass=1-T` is a choreography bookkeeping quantity, not a physical integral of the implicit field. Stroke growth also draws from an explicit authored reservoir. Dismissal creates a throat that narrows, breaks and recoils before the remaining field disappears. Events trigger modest field impulses once per site and reset on mode changes. Figure 11 shows exit at normal size; enlarged diagnostics alone would overstate its readability.



Figure 11. Throat narrowing, separation and recovery during approved M7.2 dismissal. Actual browser frames, not generated liquid artwork.

Algorithm 5 — liquid frame.

```
read idle intent; reset only choreography/input bookkeeping on state change
refresh masks only for layout or minute changes
advance bounded controller time and compute local contact/drop/neck state
enqueue each contact or break impulse once
advance persistent field through ping-pong targets
evaluate mapped glyph + implicit masses at displaced coordinates
derive artistic thickness normal; sample current clean ocean image
blend optical presentation without feeding output color back into state
```

19. Lifecycle and timing

Subsystems deliberately reject or cap suspension debt. The idle controller clears its accumulator when hidden. Its per-call duration is at most 0.05 s; the tissue path also has finite-duration guards. These are stability policies, not exact real-time continuation through a background pause.

Equation (22), **CODE-EQUIVALENT** for the idle accumulator with $h = 1/60$, is

$$A' = \min(A + \min(dt, 0.05), 3h), \quad N = \min(3, \lfloor (A' + 10^{-10})/h \rfloor), \quad A^+ = A' - Nh.$$

Hidden/invalid-input branches clear debt before this expression. The renderer is serialized to avoid concurrent frame work. Idle shader linking may continue while ordinary frames render, but the optical output is not exposed until its first valid draw. Resize changes target dimensions and invalidates field state as required; teardown waits for in-flight ownership before disposing resources.

Algorithm 6 — bounded lifecycle.

```
if hidden or invalid input: clear relevant debt/input latch; do not catch up
otherwise cap admitted duration and accumulate at most the allowed steps
run fixed steps, retaining only a fractional remainder
serialize scene/output work and restore renderer state on exceptions
publish readiness only after successful preparation
on teardown: stop listeners, settle in-flight ownership, dispose once
```

20. Implementation and artifact boundary

The locked software stack is React 19.2.0, Three.js 0.175.0 and Vite 6.4.2. The build uses the existing static client/Sites packaging path; publication does not modify its worker or deployment configuration. Scientific diagrams, plots, metadata and portable browser tools are publication additions only. The source manifest checks byte parity of runtime files against the approved commit rather than inferring parity from a successful build.

The repository contains historical modules and assets that are no longer active. Import paths decide implementation truth. In particular, the old idle renderer and old camera cannot be cited as current simply because their files remain. The provenance audit separates active scanned albedo,

archived models, generated graphics-recovery artwork, original publication captures and external reference-only material.

21. Experimental methodology

Fresh performance measurements run separately from screenshots, video capture, encoding, builds and tests. The recorded quantity is the interval between completed asynchronous ocean-frame callbacks. It is not a GPU timestamp or a measurement of shader duration. The harness records its matching method and rejects empty samples. Browser scheduling, compositor cadence, CPU work and driver submission all contribute.

The minimum scenario set is opening, midwater, bubble passage, sanctuary, Explore, settled idle and active idle manipulation. Each record includes runtime SHA, browser and backend, hardware, viewport, drawing buffer, DPR, quality/bloom state, warm-up, duration and raw intervals. A deterministic random seed makes initial conditions more comparable; real-time interactive trajectories still depend on scheduling. Existing development hooks select review states; they do not authorize changed runtime algorithms.

The fresh-results table is generated from retained data, not copied from a milestone report. Median, p95, maximum and counts above 50 ms are reported together. Adverse intervals remain in the data. A quiet repeated frame does not prove that an active transition is equally cheap. Conversely, a capture-induced stall is not silently used as evidence of ordinary runtime cost.

22. Results

22.1 Correctness and invariants

A clean dependency installation succeeded. The first pre-build suite passed 129 of 130 tests; the remaining test required generated packaging output. Building first satisfied that prerequisite, after which all 130 existing tests passed. The approved-motion check covered 720 frames and 24 exact checkpoints; the legacy default-animal parity check covered 240 frames and eight checkpoints. These narrow checks establish their stated comparisons, not biological correctness or universal visual acceptance.

22.2 Visual evidence

The figure set and S1–S12 motion gallery expose separate claims: anatomy and pulse, turn-related lag, current activation, detail transitions, bubble optics, camera modes, sanctuary behavior, and the idle lifecycle. Actual browser frames are distinguished from source-derived diagrams. Normal-size merge and pinch footage is included because a zoomed crop can make an otherwise invisible event appear convincing. Human art-direction approval remains distinct from automated correctness and from any unperformed perception study.

22.3 Fresh performance

Table 3 and Figure 12 are generated from the fresh benchmark record. They use the exact approved runtime and preserve all raw adverse samples. Measurements are frame intervals, not GPU timings. No comparative performance claim against other projects is made.

Table 3. Fresh approved-runtime frame intervals (milliseconds).

Scenario	Median	p95	Maximum	>50 ms	Frames
opening	16.70	18.60	28.80	0	1800
midwater	16.70	17.80	27.20	0	1800
bubble passage	16.70	18.70	65.30	1	1798
sanctuary	16.70	18.00	25.90	0	1800
Explore	16.70	18.50	31.80	0	1800
settled M7	16.70	17.90	26.00	0	1800
M7 manipulation	16.70	18.20	30.00	0	1803

Actual NVIDIA WebGL2; 1280×900 viewport and drawing buffer; DPR 1; bloom off. Eleven-second initial warm-up plus scene/idle warm-up; 30-second measurement per case. Existing development hooks select the approved runtime states. These are render-completion intervals, not GPU timings. Single-run observations, not confidence intervals. Bubble passage includes its natural decay. No capture or video encoding during timing.

The measured browser is Brave/Chromium 152.0.7977.76, with ANGLE OpenGL ES 3.2 on an NVIDIA RTX 4070. The Linux host uses an Intel Core i5-14600K and 62.5 GiB RAM. Its local llama service remained running consistently; this run does not isolate service contention. Existing development fixtures hold DPR at 1 without adaptive downshift. Median and p95 use sorted samples at floor(0.5n) and floor(0.95n), respectively. The bubble case retained one 65.3 ms interval; its cause is not established. The natural bubble window may end within the 30-second observation. Draw-call snapshots were not sampled in that timing run; separate diagnostic population records must not be relabeled as simultaneous benchmark draw counts. Raw arrays and environment records are retained in `results/benchmark.json`.

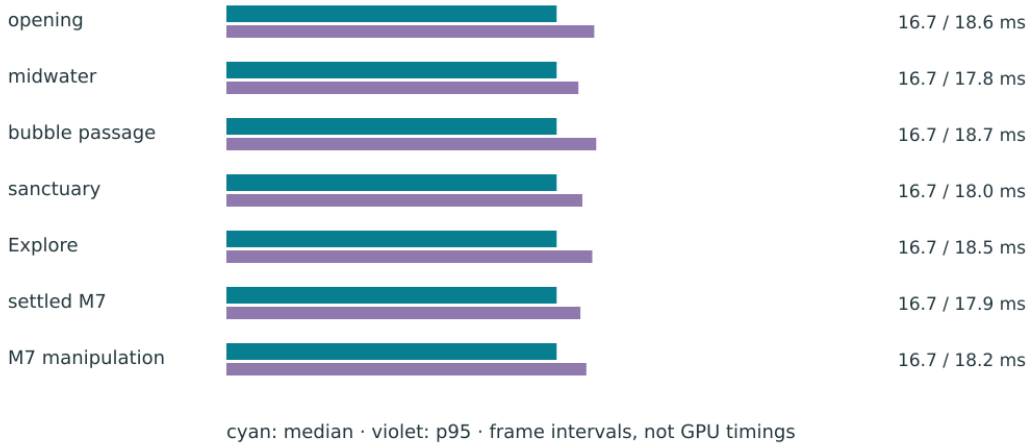
22.4 Compatibility scope

Table 4. Compatibility scope of this publication, not a general support promise.

Table 4 separates the actual backend from emulation and untested environments.

Environment	Scope
NVIDIA RTX 4070, Brave/Chromium, WebGL2	actual graphics path used for primary captures and fresh results
Narrow and 390×844 portrait viewports	browser emulation; not physical-mobile validation
Hardware WebGPU full ocean	NOT TESTED in this publication; legacy full-ocean issue remains separate
SwiftShader/software WebGPU	NOT TESTED in this publication unless an explicitly separate record is supplied
Safari, Firefox, iOS/Android hardware	NOT TESTED

Fresh approved-runtime frame intervals



PELAGIC · source-derived diagram · approved M7 bce3571

Figure 12. Fresh publication-runtime median and p95 frame intervals. Labels identify scene and measurement conditions; no screenshot/video work runs during timing.

23. Ablations and rejected approaches

The supplement distinguishes controlled same-runtime ablations from historical changes. Turning a current feature off isolates a contribution but does not necessarily provide a viable alternative implementation. A historical before/after may change several systems and cannot support a causal speedup claim without controlling those differences.

Table 5. Engineering questions and evidence interpretation.

Table 5 distinguishes the frozen diagnostic, a time course and historical comparisons.

Evidence	Observation / question	Limit
Frozen hero-refraction on/off, Figure 8	live image displacement; ambient films retained	not physically correct multilayer refraction
Pointer, rest and recovery, S10	time-course observation of remembered deformation	not an isolated solver ablation or conserved liquid
Historical camera correction vs bounded path, supplement N1	rationale for simpler ownership	not an isolated benchmark speedup
Historical separate idle renderer vs shared M7, supplement N4	architecture and source ownership	not universal browser compatibility

A connected-particulate visible/hidden ablation was not run for this publication; S3 is an activation time course, not that controlled comparison. The frozen optical pair has SSIM about 0.969, a descriptive image difference only: it is neither a visual-quality score nor evidence of physically correct refraction. The comparison uses the existing review interface, not edited runtime source.

Rejected giant dark lenses, bright repetitive geology and early camera corrections are presented as development history, not as straw-man competitors. The later Asset2 experiment missed requested image-metric improvements and its optional GTAO path encountered unhelpful depth sampling. It is not promoted to approved runtime or used as proof that ambient occlusion improved the artwork. The input-boundary defect behind an earlier idle failure is also distinguished from a shader diagnosis: malformed synthetic pointer data could poison state, so validation had to identify the first invalid input rather than sanitize final geometry.

24. Limitations

Pelagic is not calibrated to a species, a measured ocean volume, or physical tissue properties. Its bell and folds are procedural art direction. Constraint iterations and clearance heuristics can reduce, but not eliminate, intersections. The appendages do not solve continuum elasticity or complete self-collision. Route separation is not a validated collective-behavior model.

Current, wake and plume fields are bounded approximations. Their shared timing and spatial dependence improve coherence, but there is no fluid–structure coupling, pressure solve, mass conservation, thermal transport or prediction of real vent behavior. Distance-dependent water appearance is not a validated participating-media reconstruction. Geological textures contribute surface detail without converting authored lighting into a measured BRDF.

Image-space optics have finite information. They cannot recover offscreen or occluded radiance. Transparent tissue lacks its own resolved depth layer. Virtual planes, displacement caps, grazing fades and softened indices are deliberate compromises. Multiple optical effects share a clean source rather than recursively transporting light. The pinned WebGL depth-resolution issue further limits occlusion confidence; attachment presence must not be advertised as correct depth behavior.

The liquid clock is a two-dimensional scalar presentation with authored contact timing and small persistent deformation. Its necks and reservoirs do not prove conserved volume. Resize may reset local field state. Hidden-tab time is discarded, not physically simulated. Preparation can still impose one-time driver work; steady-state frame intervals do not erase that cost.

The hardware matrix is narrow. Results on one NVIDIA desktop and emulated viewports do not establish mobile performance, Safari compatibility, or a fixed frame budget on all machines. Visual approval is Mani’s artistic judgment, not a controlled user study. No peer-review or priority claim accompanies this independent technical preprint.

25. Reproducibility

Use the recorded runtime commit, not an unspecified current main branch. Install the lockfile, build, run tests and parity checks, then start the local review server. `REPRODUCIBILITY.md` gives portable commands and tool versions. The publication browser harness records scenario state and metadata separately from video. Its benchmark mode never captures frames. Large source clips and raw capture material remain outside normal Git; curated figures, lightweight GIFs and numerical records fit the publication media budget.

The release staging directory contains the paper and supplement PDFs, a reproducibility archive, supplemental motion archive, prepared arXiv source and SHA-256 manifest. These are local deliv-

erables, not evidence of an issued GitHub release, DOI or arXiv submission. The currently public ocean is linked for visitors with an explicit version caveat. Publication requires a separate author approval step.

26. AI assistance and provenance

Mani Marami Milani directed and reviewed the artwork. Codex/OpenAI models assisted with implementation, source auditing, documentation, test tooling and publication preparation. This disclosure is not an independent verification of every generated statement; the source map, measured data and human review gate remain necessary. Existing generated artwork is used for graphics recovery or reduced-motion fallback, not substituted for successful runtime motion evidence.

Original code is MIT licensed; original publication text, diagrams and project-generated screenshots/media are CC BY 4.0. Third-party code, fonts, textures and archived models retain their own terms. This policy does not relicense external research material. Scientific figures from other authors are linked or cited, not copied into the publication’s original figure set.

27. Conclusion

Pelagic’s useful result is an inspectable connection between an organism, its surrounding water and an optical interface. Shared state, bounded time, consistent coordinate spaces and explicit resource ownership matter as much as individual shading techniques. The approved system preserves a procedural animal through motion and detail changes, lets its environment respond, and uses the real moving ocean as the source for liquid optics.

The artifact makes those decisions reproducible without overstating them. Its strongest claims are source-level integration, demonstrated behavior and measured performance under stated conditions. Its physical and compatibility limits remain visible. Further biological calibration, more complete optical transport and broader hardware validation would be separate work, not implied achievements of this publication.

References

- Blinn, James F. 1982. “A Generalization of Algebraic Surface Drawing.” *ACM Transactions on Graphics* 1 (3): 235–56. <https://doi.org/10.1145/357306.357310>.
- chiuhans111. n.d. *FluidGlass*. <https://github.com/chiuhans111/fluidglass>.
- Costello, John H., Sean P. Colin, John O. Dabiri, Brad J. Gemmell, Kelsey N. Lucas, and Kelly R. Sutherland. 2021. “The Hydrodynamics of Jellyfish Swimming.” *Annual Review of Marine Science* 13: 375–96. <https://doi.org/10.1146/annurev-marine-031120-091442>.
- Gemmell, Brad J., John H. Costello, Sean P. Colin, et al. 2013. “Passive Energy Recapture in Jellyfish Contributes to Propulsive Advantage over Other Metazoans.” *Proceedings of the National Academy of Sciences* 110 (44): 17904–9. <https://doi.org/10.1073/pnas.1306983110>.

- Müller, Matthias, Bruno Heidelberger, Marcus Hennix, and John Ratcliff. 2007. “Position Based Dynamics.” *Journal of Visual Communication and Image Representation* 18 (2): 109–18. <https://doi.org/10.1016/j.jvcir.2007.01.005>.
- Niehus, Niklas. n.d. *Aurelia*. <https://github.com/holtsetio/aurelia>.
- Poly Haven. n.d. *Rock 07*. https://polyhaven.com/a/rock_07.
- Three.js contributors. 2025. *Three.js r175 Source and Rendering Implementation*. <https://github.com/mrdoob/three.js/tree/r175>.
- Wyman, Chris. 2005. “An Approximate Image-Space Approach for Interactive Refraction.” *ACM Transactions on Graphics* 24 (3): 1050–53. <https://doi.org/10.1145/1073204.1073310>.

Appendix A. Reading the artifact

`source-map.md` connects every equation to the fixed runtime. `notation.md` defines symbols and spaces. The claim ledger separates source proof, benchmark proof, observation and heuristics. The supplement contains motion descriptions, extended measurements, ablations and historical negative results. The manifest records file hashes and distinguishes runtime SHA from publication SHA.

Figures 1, 9–11 are runtime evidence; Figures 2–8 include original explanatory diagrams or source-derived plots. Figures 4, 6 and 8 also include labeled browser frames, as their captions state. Figure 12 is generated from fresh raw measurements. No diagram is represented as a rendered result.